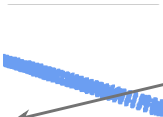


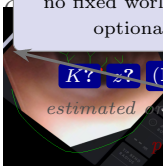
Inputs (any)



FreeReg: free-coordinate map

$$\mathbf{x}_p = \pi_D(K \mathbf{T}_{cm}(\mathbf{X}_m + W(\mathbf{X}_m)))$$

no fixed world frame; each factor
optional & degradable

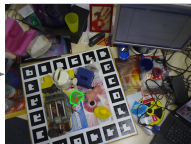


$K?$ $z?$ $(\mathbf{R}, \mathbf{t})?$ $W?$

estimated or clamped

photometric residual = online validation

Output



anchored AR / MR overlay

image / video

calibration-free observability-bound occlusion-aware